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VRLA and Tubular batteries

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tion is very small with respect to the total fault current.

3. Another mode that uses feedback control with P+I mode has also been developed and tested. However, it was not given in this circuit as it was not required in this small rating of 5A. In that case, the 9V-0-9V transformer is to be connected to the output. The given program is called 'feedforward' because it senses the input and corrects by adding or subtracting.

The feedback program senses the output and controls the pulse-width by noting the error and also evaluating the error integral and using both to adjust the change in pulse-width for every update. Increasing the amount of integral action makes the output voltage hunt. The parameters are very much dependent on the transformer designed. So one has to vary the integral action amount added to the proportional action in such a manner that there is no swinging—response to the input change is good enough.

4. The reader's understanding of the switching is wrong. Both the transistors conduct in both boost and buck mode. Consider boost mode: For one half cycle of the mains supply, one MOSFET (let's call it 'A') conducts, and for the other half cycle, the second MOSFET (let's call it 'B') conducts. If it is to buck, the order changes to B first and then A. In this manner, the phase of the output changes by 180 degrees between boost and buck. This automatic method of changing the voltage to add/subtract is a new idea introduced in this circuit.

The circuit has been published only after testing its working at the EFY lab. One ampere on the primary side of the buck/boost transformer can cope up for 5A on its secondary winding, because of the voltage ratio 250:50. Thus a 1A diode is sufficient for 5A rating. The diodes can be changed to 5408 for higher rating.

5. When the input voltage goes low, the secondary output of the 9V-0-9V transformer falls equally. But even this low a voltage is sufficient for operating the 7805. A 10V-0-10V transformer can be used. In that case, the zener diode should be increased

in voltage accordingly (8.2V to 9.1V+).

A reverse-connected diode can be placed across gate to source of the MOSFET. But it increases the capacitance at gate to source and the switching is not that fast. There is no negative voltage on the pulse drive from the ULN2003. The ULN provides a positive-going pulse, which swings from 0V to the full positive supply (9V). Since this path to gate has no inductance, there is no likelihood of any ringing which alone can bring about a negative voltage on the gate.

PWM mode operation for voltage generation is a novel concept introduced in this design. Normally, PWM pulses are used only to generate current flow in an inductive circuit, such as a motor winding.

Twilight Using White LEDs

The 'Twilight Using White LEDs' circuit published in April issue cannot possibly work as the input voltage is 12V and the characteristic voltage across five white LEDs is $3.6V \times 5 = 18V$.

Colin Mitchell
Melbourne, Australia

The author Pradeep G replies:

I guess you have used high-wattage LEDs. I tried my circuit with low-current, bright LEDs. There is no error in the circuit. For more brightness, avoid one LED in each row. Supply voltage is not critical. 12V-18V DC can be applied.

Triple-mode Tone Generator

In the 'Triple-mode Tone Generator' circuit published in July issue, the author says that the output current of IC 556 is 200 mA. This output is amplified using a BC547 transistor. But BC547 is a small-signal transistor with a current capacity of about 100 mA only. The circuit should use a medium-power transistor like SL100 or BD139 in place of BC547.

Pradeep G
Bangalore.

The author Ashok K. Doctor replies:

I thank Pradeep G for pointing out the mistake. Yes, apart from SL100 and BD139, you can also use BC337, 2N2222A or equivalent in place of BC547.